

Employment

- 6/2024 – **Doctoral Faculty**, *CUNY Graduate Center*
Present
- 8/2022 – **Assistant Professor**, *CUNY Baruch College*
Present
- 9/2019 – **Postdoctoral Assistant Professor**, *University of Michigan*
7/2022 Supervisor: Stephen DeBacker
- 8/2018 – **Postdoctoral Fellow**, *University of Alberta*
7/2019 Supervisor: Manish Patnaik
- 9/2014 – **NSF Postdoctoral Fellow**, *University of Toronto*
7/2018 Sponsoring Scientist: Florian Herzig
- 3/2016 – **EPDI Postdoctoral Fellow**, *Max Planck Institute for Mathematics*
8/2016 Supervisor: Geordie Williamson
- 9/2015 – **EPDI Postdoctoral Fellow**, *Institut des Hautes Etudes Scientifiques*
2/2016 Supervisor: Ahmed Abbès

Education

- 9/2009 – **Ph.D. in Mathematics**, *Columbia University*
5/2014 Advisor: Rachel Ollivier
- 9/2005 – **B.A. in Mathematics (Honors), Minor in Music**, *New York University*
5/2009 Magna Cum Laude, Phi Beta Kappa

Awards/Honors/Funding

- 2023 – 2026 PSC-CUNY Traditional B Research Award ($\times 3$), \$6,000 each
- 2021 – 2024 NSF Standard Grant [DMS-2101836](#) (transferred to [DMS-2310225](#)), \$124,050
- 2019 U(M) Math Department Holiday Party Baking Contest – 1st place
- 2019 – AIM SQuaRE Funding
- 2019 BIRS Research in Teams
- 2019 Funding to attend workshop “Hot Topics: Recent progress in Langlands Program” at MSRI
- 2017 Université Montpellier Travel Grant
- 2015 – 2016 EPDI Fellowship
- 2014 – 2018 NSF Mathematical Sciences Postdoctoral Research Fellowship, \$150,000
- 2013 Funding to attend summer school “New Geometric Techniques in Number Theory” at MSRI
- 2010 NSF Graduate Research Fellowship Honorable Mention
- 2009 – 2014 Faculty Fellow/Teaching Fellow in the Department of Mathematics, Columbia University
- 2009 – 2014 Funded by NSF RTG Grant DMS-0739400 in Number Theory, Columbia University
- 2009 Hollis Cooley Memorial Prize for Excellence in Mathematics, New York University

Publications and preprints

- *To be or not to be local*
(with Christophe Breuil, Florian Herzig, Yongquan Hu, Stefano Morra, Benjamin Schraen, and Sug Woo Shin)
Preprint
- *Mod p Iwasawa algebras of pro- p Iwahori subgroups*
(with Rudy Ariaz, Steven Creech, Bryan Hu, Simran Khunger, Bharatha Rankothge, and Bobby Zixuan Zhang)
Preprint
- *Parahoric Hecke Ext-algebras in characteristic p*
(with Rachel Ollivier and Jacob Stockton, and a joint appendix with David Schwein)
To appear in Transactions of the AMS
- *Derived Satake morphisms for p -small weights in characteristic p*
(with Cédric Pépin)
Math. Ann. **392** (2025), no. 4, 5725–5785.
- *Gelfand–Kirillov dimension for mod p representations of p -adic unitary groups of rank 2*
(with Stefano Morra)
Representation Theory **29** (2025), 893–967.
- *Derived right adjoints of parabolic induction: an example*
Pacific J. Math. **321** (2022), no. 2, 345–358.
- *Functorial properties of pro- p -Iwahori cohomology*
J. London Math. Soc. (2) **104** (2021) 1572–1614.
- *On the existence of admissible supersingular representations of p -adic reductive groups*
(with Florian Herzig and Marie-France Vignéras, and an appendix by Sug Woo Shin)
Forum of Mathematics, Sigma **8** (2020), e2.
- *Serre weight conjectures for p -adic unitary groups of rank 2*
(with Stefano Morra)
Algebra & Number Theory **16** (2022), no. 9, 2005–2097.
- *The first pro- p -Iwahori cohomology of mod- p principal series for p -adic GL_n*
Trans. Amer. Math. Soc. **372** (2019), 1237–1288.
- *Hecke module structure on first and top pro- p -Iwahori cohomology*
Acta Arithmetica **186** (2018), 349–376.
- *Irreducible admissible mod- p representations of metaplectic groups*
(with Laura Peskin)
Manuscripta Math. **155** (2018), no. 3–4, 539–577.
- *Homological dimension of simple pro- p -Iwahori–Hecke modules*
Math. Res. Lett. **26** (2019), no. 3, 769–804.
- *Pro- p -Iwahori invariants for SL_2 and L -packets of Hecke modules*
International Mathematics Research Notices (2016), no. 4, 1090–1125.
- *A classification of the irreducible mod- p representations of $U(1, 1)(\mathbb{Q}_{p^2}/\mathbb{Q}_p)$*
Annales de l’Institut Fourier **66** (2016), no. 4, 1545–1582.
- *Hecke modules and supersingular representations of $U(2, 1)$*
(with Peng Xu)
Representation Theory **19** (2015), 56–93.

Seminar Talks

- 5/2025 **Number Theory Seminar**, POSTECH-PMI
- 12/2024 **Number/Representation Theory Seminar**, University of Toronto
- 3/2024 **Automorphic Forms and Representation Theory Seminar**, Purdue University
- 11/2023 **Philadelphia Area Number Theory Seminar**
- 10/2023 **Colloquium**, Southern Illinois University
- 11/2022 **Automorphic Forms and Representation Theory Seminar**, Purdue University
- 5/2022 **Iwahori–Hecke algebras and the mod- p Langlands program (4-part lectures series)**, Université Sorbonne Paris Nord (Paris XIII)
- 3/2022 **Number Theory Seminar**, Penn State
- 11/2021 **RTG Seminar**, University of Michigan
- 10/2021 **Joint Number Theory Seminar**, Columbia–CUNY–NYU
- 11/2020 **RTG Seminar**, University of Michigan
- 10/2020 **Number Theory Seminar**, Harvard
- 8/2020 **New Developments in Number Theory**, hosted by POINT, online
- 2/2020 **Topology, Analysis & Geometry Seminar**, Central Michigan University
- 11/2019 **Algebra, Combinatorics, and Geometry Seminar**, University of Pittsburgh
- 10/2019 **Algebra and Number Theory Seminar**, University of Arizona
- 4/2019 **Number Theory Seminar**, Northwestern
- 2/2019 **Research Presentation**, Trinity College Dublin
- 2/2019 **Number Theory Seminar**, University of British Columbia
- 1/2019 **Number Theory/Representation Theory Seminar**, University of Alberta
- 4/2018 **Number Theory Seminar**, UCLA
- 4/2018 **Number Theory Seminar**, UC San Diego
- 4/2018 **Number Theory Seminar**, Caltech
- 10/2017 **Séminaire de Géométrie Algébrique**, Université Montpellier 2
- 9/2016 **Number/Representation Theory Seminar**, University of Toronto
- 7/2016 **Séminaire de Théorie des Nombres**, Université Montpellier 2
- 6/2016 **Mittagsseminar zur Arithmetik**, Universität Münster
- 4/2016 **Oberseminar**, Max Planck Institute for Mathematics
- 11/2015 **Séminaire Groupes Réductifs et Formes Automorphes**, Institut de Mathématiques de Jussieu
- 11/2015 **Postdoc Seminar**, IHES
- 3/2015 **Number/Representation Theory Seminar**, University of Toronto
- 3/2015 **Arithmetic Geometry Seminar**, McMaster University
- 4/2014 **Princeton/IAS Junior Faculty Number Theory Seminar**, Princeton University
- 4/2014 **Group, Lie and Number Theory Seminar**, University of Michigan
- 2/2014 **Representation Theory Seminar**, CUNY Graduate Center
- 10/2013 **Number Theory Seminar**, University of British Columbia
- 6/2013 **Séminaire Groupes Réductifs et Formes Automorphes**, Institut de Mathématiques de Jussieu

Conference Talks

- 11/2025 **Arithmetic Geometry in Shenzhen**, Shenzhen
- 10/2024 **AMS Sectional Meeting**, SUNY Albany
- 9/2024 **Iwahori Ext-algebras and mod- p representations of p -adic groups (mini-course)**, IIT Jammu
- 6/2023 **CMS Summer Meeting**, University of Ottawa
- 5/2023 **Hecke Algebras as a Unifying Theme**, University of British Columbia
- 5/2023 **Representation Theory, L-functions, and Arithmetic**, University of Pittsburgh
- 6/2022 **Journées Arithmétiques du LAGA**, Université Sorbonne Paris Nord (Paris XIII)
- 5/2022 **AMS Sectional Meeting**, University of Denver
- 3/2022 **Midwest Representation Theory Conference**, University of Michigan
- 7/2021 **Mathematical Congress of the Americas**, Buenos Aires, Argentina (online)
- 8/2019 **Nonassociative Algebras and Geometry**, Bonne Bay Marine Station
- 6/2019 **CMS Summer Meeting**, Regina (unable to attend)
- 5/2019 **Padova school on Serre conjectures and the p -adic Langlands program**, Padova
- 5/2019 **Alberta Number Theory Days XI**, BIRS
- 10/2018 **AMS Sectional Meeting**, University of Michigan
- 4/2018 **Upstate Number Theory Conference**, SUNY Buffalo
- 4/2017 **AMS Sectional Meeting**, University of Indiana, Bloomington
- 6/2016 **Geometric methods in the mod p local Langlands correspondence**, Centro di Ricerca Matematica Ennio De Giorgi
- 12/2015 **CMS Winter Meeting**, Montreal (unable to attend)
- 3/2014 **Conference on Modular Iwahori-Hecke Algebras**, Humboldt-Universität zu Berlin
- 12/2013 **Local Langlands Conjecture and Galois Representations**, Tata Institute
- 10/2013 **AMS Sectional Meeting**, Temple University
- 5/2013 **Hecke algebras in Number Theory and Categorification**, Columbia University
- 4/2013 **Modular Representation Theory of Finite and p -adic Groups**, IMS Singapore

Teaching

- **Baruch College:**
 - Elements of Calculus I (S23)
 - Calculus II (F22, F23, F25)
 - Elementary Probability (S25)
 - Linear Algebra (S24, F25)
 - Mathematical Analysis (F22)
 - Graph Theory (S25)
 - Introduction to Probability (S23, S24)

- **University of Michigan:**
 - Calculus I (F19)
 - Linear Algebra (F21)
 - Linear Algebra – IBL (W21)
 - Applied Modern Algebra (F19, F20)
 - Elementary Number Theory (W20, W21)
 - Number Theory Lab (W20)
 - Probability Theory (W22)
 - Representation Theory of p -adic Groups (Graduate) (F20)
- **University of Alberta:**
 - Calculus I (F18)
- **University of Toronto:**
 - Linear Algebra I (F16)
 - Linear Algebra II (W17, Sum17)
- **Columbia University:**
 - Calculus III (Sum11)
 - Calculus IV (Sum12, Sum14)

Outreach

- **CUNY Baruch College:**
 - mentored a high school student research project about continued fractions and fundamental units
 - mentored a high school student research project about elliptic curves
- **University of Michigan:**
 - participated in Super Saturdays for the [Ypsilanti Math Corps](#)
 - gave talks at Michigan Math Circle, U(M) Undergraduate Math Club
- **University of Alberta:**
 - Science Fair Judge for middle school students at Aurora Academic Charter School
 - participated in weekly JAMES (Junior Alberta Math for Eager Students) Math Circle
- **University of Toronto:**
 - gave numerous presentations to high school students about algebraic topology, modular arithmetic, Arrow's theorem, the stable marriage problem, random walks, partial differential equations, and Fermat's theorem on two squares
 - mentored high school student Behshad Jahanpour's research project about p -adic numbers
- **Columbia University:**
 - gave presentations to Undergraduate Math Society about p -adic numbers and mathematical puzzles

- **New York University:**

- gave presentations to high school students about modular arithmetic, algebraic topology, and elliptic curves at cSplash (one day mathematics outreach event at NYU)
- co-organizer of cSplash, 2007 – 2009

Service

Referee/quick opinion for: *Advances in Mathematics*, *Algebra and Number Theory*, *American Journal of Math*, *Bulletin of the LMS* ($\times 2$), *Compositio Mathematica*, *Duke Math Journal*, *Forum Mathematicum*, *IMRN*, *International Journal of Number Theory* ($\times 2$), *Israel Journal of Math*, *Journal of Algebra*, *Journal de Théorie des Nombres de Bordeaux* ($\times 2$), *Journal of Number Theory*, *Journal of the EMS*, *Journal of the LMS* ($\times 2$), *Manuscripta Mathematica* ($\times 2$), *Math Annalen*, *Math Research Letters*, *Math Zeitschrift*, *Proceedings of the AMS*, *Proceedings of the LMS*, *Representation Theory* ($\times 8$), *Transactions of the AMS*

- 2025 Co-organizer of special session on *Automorphic Forms and Representation Theory: current trends* (with A. Raghuram), AMS Sectional Meeting, Tulane University
- 2025 Problem Session Leader, [Arizona Winter School 2025: Representation theory of \$p\$ -adic groups](#) (included compiling [exercise sheet](#))
- 2024 Project Leader, [Rethinking Number Theory](#)
- 2023 Research mentor for 3 undergraduate students at CUNY Baruch
- 2023 – Co-organizer of [Baruch College Distinguished Lecture Series](#) (with Adam Sheffer)
- 2023 – Proposal reviewer for PSC CUNY Grant
- 2023 Co-organizer of special session on [Homological Aspects of \$p\$ -adic Groups and Automorphic Representations](#) (with A. Raghuram), AMS Sectional Meeting, University at Buffalo
- 2023 – Putnam exam training sessions, Baruch College
- 2022 Summer REU co-mentor for Gahl Shemy at the University of Michigan (with Charlotte Chan)
- 2022 Thesis committee member for Yiwang Chen
- 2021 – 2022 Co-organizer of RTG seminar on Representation Theory (with Charlotte Chan)
- 2021 Co-organizer of [Michigan Algebra and Number Theory Intercity Symposium \(MANTIS\)](#) (with Yuan Liu)
- 2021 Summer REU co-mentor for Drew Keisling and Dylan Pentland at the University of Michigan (with Tasho Kaletha)
- 2021 Co-organizer of special session on [Representations of \$p\$ -adic groups and Langlands correspondences](#) (with Monica Nevins), 2021 CMS Summer Meeting, Ottawa.
- 2021 Panelist for session on applying to academic jobs for graduate students at University of Michigan
- 2020 – 2023 Co-organizer of [POINT: New Developments in Number Theory](#)
- 2020 Summer REU mentor for Alexander Wang at the University of Michigan
- 2020 Co-organizer of the [Cross Atlantic Representation Theory and Other topics ONLINE \(CARTOON\) conference](#) (with Jessica Fintzen) — online conference with over 400 registrants
- 2019 – 2020 Co-organizer for Group, Lie and Number Theory Seminar at the University of Michigan
- 2019 – 2022 Mentor, Postdoc-Grad Student Mentorship Program, University of Michigan
- 2019 Co-organizer for Number Theory/Representation Theory Seminar at the University of Alberta
- 2016 – 2018 Co-organizer for Number Theory/Representation Theory Seminar at the University of Toronto

- 2013 Transcribed notes for Florian Herzig's Mini-Course " *p -modular Representations of p -adic Groups*" at the summer school "Modular Representation Theory of Finite and p -adic Groups"
- 2012 – 2014 Departmental graduate student representative in Mathematics Department at Columbia University
- 2011 – 2014 Organized/co-organized student number theory seminars on Deligne–Lusztig theory, theta correspondence, deformation theory, and perfectoid spaces at Columbia University
- 2011 Co-organizer of Summer REU on Apollonian Circle Packings (with Tim Heath and Ian Whitehead)

Languages

English, Polish